Biodiversity

AI/ML based model to mitigate the declining population of wild pollinators due to climate change

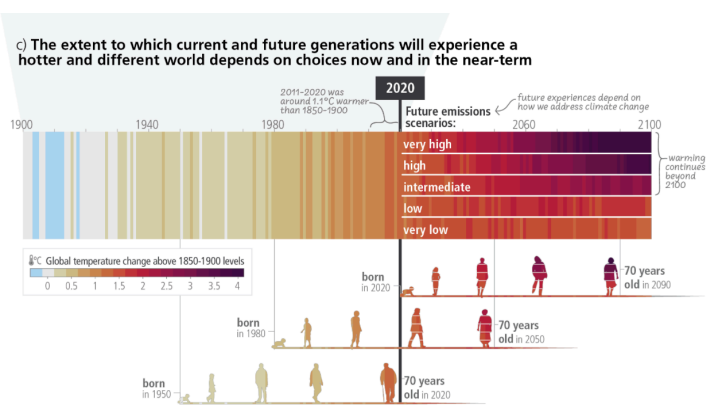
By SE23UARI137, SE23UARI169, SE23UECM070, SE23UARI109, SE23UCAM019, SE23UCSE137, SE23UARI025, SE23UARI099, SE23UMEE023, SE23UMEC027.

Abstract

Climate change poses a serious threat to biodiversity, especially wild pollinators that are vital for ecosystems and agriculture. This report presents an AI-driven solution that combines ecological and climate data to monitor pollinator populations, predict risks, and plan conservation efforts. By using advanced techniques like CNNs and ARIMA, the approach aims to restore habitats, support food security, and promote global sustainability in response to the challenges of pollinator decline.

1 Introduction

In recent years climate change has emerged as one of the most important issues we face as the gradual rise in temperature, rise in sea levels, warm currents negatively affect our world and its inhabitants. This in turn affects our world's biodiversity and agricultural productivity, due to a decline in the population of wild pollinators. Pollinators like bees butterflies and birds face problems like rising temperature, altered precipitation, late or early blooming of flowers which then leads to a mismatch between plants blooming and pollinator activity, habitat loss, and increased vulnerability to diseases.



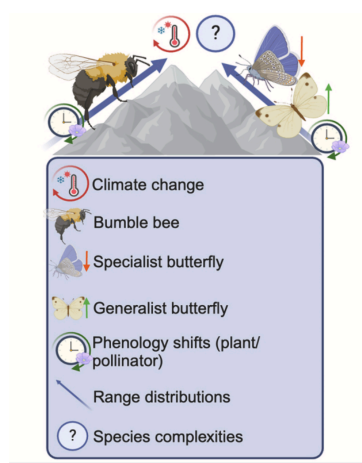
`figure` - illustrates the rising temperatures and highlights the urgent need for action to protect future generations.[1]

2 Impact of Climate Change on Pollinators and Biodiversity

Climate change directly affects pollinator populations, all of which are crucial for maintaining biodiversity and food security. Let's understand primary climate-related issues on pollinators :

2.1 Shifting in timing (Phenology):

Rise in temperature, altered weather patterns, and various other climatic conditions creates discrepancies between the timing of plants flowers and pollinating activity which means the flowers are either blooming early or late when the pollinators like bees, butterflies, and other insects are not present resulting in reduced reproductive success in many species.



`figure`[2]

[1] https://www.ipcc.ch/report/ar6/syr/figures/figure-spm-1

2.2 Habitat loss :

Climate change affects ecosystems worldwide. Issues like desertification, fragment pollinator habitats, makes it hard for pollinators to survive. Deforestation is decreasing the availability of food for pollinators, and changes in plant distribution are directly affecting them.

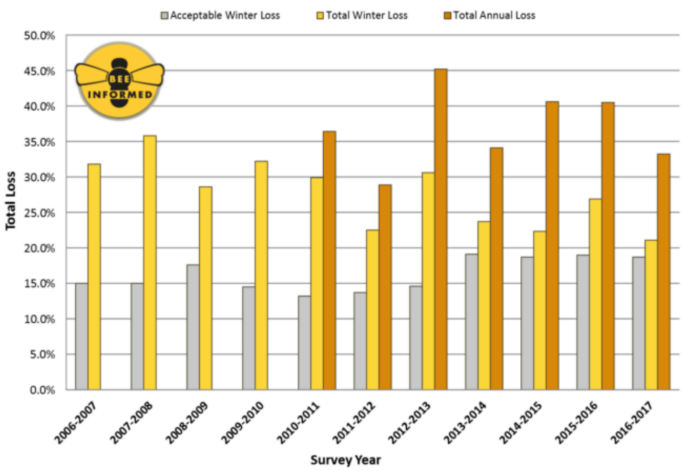


figure - Total US managed honey bee colonies Loss Estimates[3]

- The western monarch butterfly is a good example of a pollinator butterfly and here is the data on its decline in population due to various reasons mainly being climate change and habitat loss.

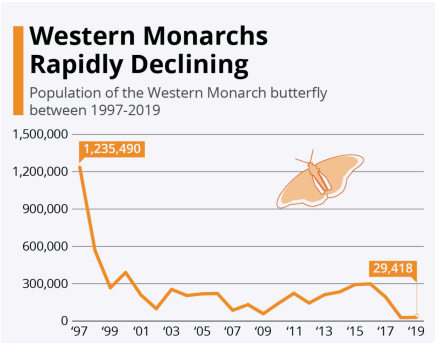


figure - Data collected during Xerces Society's annual Western Monarch Thanksgiving Count Source: Xerces Society[4]

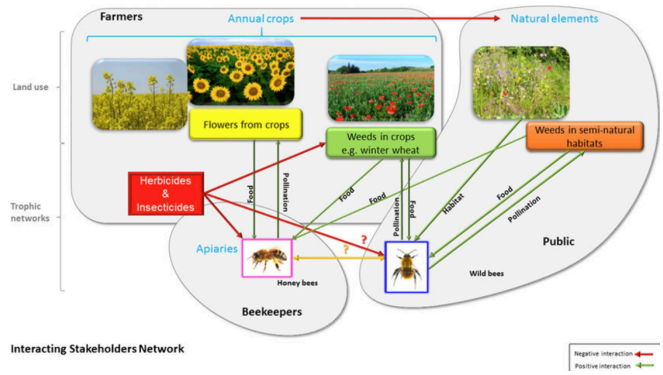
From the data above, we can infer that climate change is affecting our pollinators, which is indirectly affecting their biodiversity. They play a major role in increasing flora by helping in the reproduction of plants. Hence, one must understand how to tackle the decrease in population of pollinators due to climate change.

2.3 Plant reproduction[5] :

Pollination occurs when pollen grains are moved between two flowers of the same species by wind or animals. Successful pollination results in the production of healthy fruit and fertile seeds, allowing plants to reproduce. Nearly 90% of wild flowering plants depend on pollinators and a loss in pollinator population can disrupt the ecosystem and biodiversity on multiple levels.

2.4 Food Web[6] :

Animal - pollinated crops play an important role in global food systems. Approximately 75% of global food crops benefit from animal pollination including fruits, vegetables, nuts, and seeds. Pollinator activity directly influences both the quantity and quality of agricultural produce. For instance, apples and strawberries pollinated by insects tend to be larger, more uniform, and better tasting.

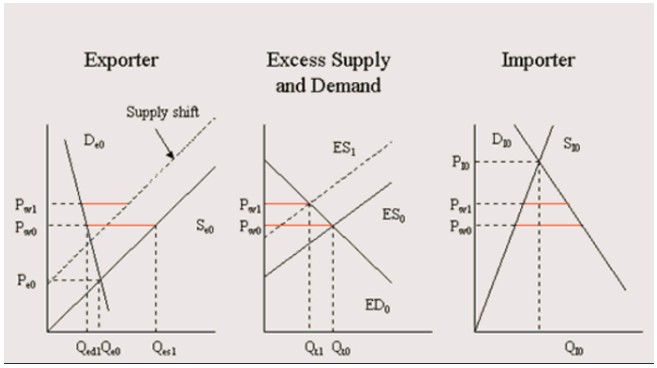


2.5 Economic impact[7]:

The decline in the population of pollinators can have economic impacts on both producers and consumers. Consumers will suffer as the commodity prices will increase due to less availability of crops. Without pollinators, more than 39 percent of crops would see a decline in their production. There is evidence supporting the future pollinator shortages affecting the economy

[2]https://pmc.ncbi.nlm.nih.gov/articles/PMC10590815/
[3]https://toolkit.climate.gov/image/2571
[4]https://www.statista.com/chart/20716/monarch-butterfly-population-decline/
[5]https://www.sciencedirect.com/science/article/abs/pii/S0167880920302929
[6]https://ourworldindata.org/pollinator-dependence
[7]https://www.cambridge.org/core/journals/agricultural-and-resource-economics-review/article/abs/economic-consequences-of-pollinator-declines-a-synthesis/88F27F7491BAA6C831161992D62BA447

of both direct crop sectors and indirectly the non-crop sectors.



`figure` - Impact of pollinator deficit[8]

A pollinator deficit in the exporting country will cause the supply curve to shift to the left. This in turn causes the excess supply curve to shift to the left creating a new equilibrium trade position.

So according to our research and a lot of Googling, we found that there is less data available on the topic we are dealing with or the data is pretty old so proposing and creating a pollinator dataset related to pollinators and their behavior towards climate change will be highly valuable.

3 Data

According to the research we have conducted, we found that there is sufficient data on monitoring pollinator populations. However, most of this data seems to be in regards to pollinators like bees, and less of other wild pollinators like hummingbirds, bats, and butterflies. We also require environmental data in order to understand how climate change specifically affects different pollinator species, so we plan to combine both climatic data and pollinator data to form a model. Many of the datasets are also region specific, hence cannot be applied globally, indicating the need for a large scale dataset that covers the globe. The pollinator datasets which we found were :

3.1 The Bumblebee Conservation Trust (BBCT) data[9] :

The Bumblebee Conservation Trust (BBCT) dataset provides valuable data on the bumblebee population in the UK. It includes various details about bumblebee species, their distribution, and population trends over time. This data is crucial for studying how climate change and environmental shifts impact pollinators, particularly bumblebees. Researchers can utilize this dataset to analyze the relationship between climate variables and pollinator behavior, helping to inform conservation efforts and biodiversity strategies.

3.2 USDA National Agricultural Statistics Service (NASS) - Pollinator Data :

This dataset collects data on the use of pollinators in agriculture including trends in pollinator number and effects of climate change on them.[10]

3.3 European Pollinator Initiative (EPI) - Pollinator Data[11] :

The EPI provides the data related to pollinator health and the biodiversity in Europe. It also includes information on now climate change an environmental changes affect pollinator population and distribution. Their main objectives are to **1)** Improve knowledge of pollinator decline its causes and consequences, **2)** Improve pollinator conservation and tackle the causes of pollinator their decline, and **3)** Mobilize society and promote strategic planning and cooperation at all levels

The major problem :

Most of these datasets only cover the data related to bees. We know that bees play a major part in pollination approximately 80% but there are other species like humming birds, butterfly, ants which also take part in pollination. Most of the data from these data sets are not accurately measured and are region specific and can't be applied in other parts of the word so the requirement for a larger dataset is a good approach to start and tackle our major problem.

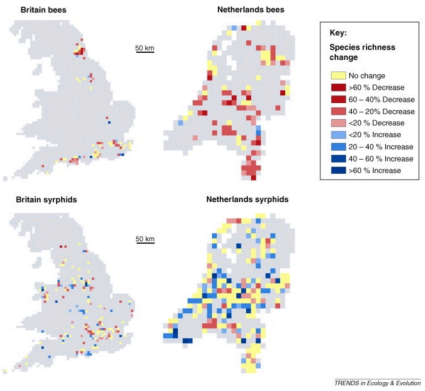
4 Research And Literature Review

Extensive literature reviews and research papers have been made on the decline of natural pollinators, and some have even proposed solutions. The following case study highlights how pollinator populations are on the decline.

[8] https://www.ipcc.ch/report/ar6/syr/figures/figure-spm-1
[9] https://www.bumblebeeconservation.org
[10] https://www.nass.usda.gov/Research_and_Science/index.php
[11] https://environment.ec.europa.eu/topics/nature-and-biodiversity/pollinators_en
[12] https://www.cell.com/trends/ecology-evolution/fulltext/S0169-5347(10)00036-4

4.1 Case study[12]

There has been a rapid decline in wild bees and hoverflies in the UK and Holland, this has in turn resulted in parallel losses in wild plant species, indicating a strong correlation between the population of pollinators and the population of wild flowers, as the flowers lack necessary pollinating derives due to the decreasing number of bees, or that the bee's population is declining due to a lack of flowers. Whatever the case, it is clear that both flowering plants and pollinators share a sensitivity to environmental changes.



Currently, there exists a few solutions to the problem of declining pollinator populations such as robot bees, precision beekeeping systems, and the World Bee Project.

Robobees refer to the machines which artificially pollinate flowers such as ground vehicles, arial drones or even flying micro robots. These robots aim to combat the declining bee population by taking on the role of a pollinator. However, they are currently not a feasible solution.

According to a study conducted by Indian farming robobees are not as efficient as natural pollinators. There exists 350,000 species of flowering plants, and an even great amount of diverse wild pollinators. With our current technology, we will not be able to effectively make a global difference in replacing natural pollinators, as we cannot meet their efficiency. Additionally, it is not an economically feasible solution due to the high development and manufacturing costs, along with the cost of deployment and maintenance. These robobees are also susceptible to harsh weather conditions such as rainstorms, hail, and snowfall, increasing repair costs.

4.2 Precision beehives

> Precision beekeeping is defined as the "strategy for the management of an apiary based on monitoring individual honeybee colonies to minimize resource consumption and maximize productivity."[13]

It utilizes various sensors in order to monitor the weight of the hive, internal temperature, humidity, and even sound. These sensors help beekeepers understand the physiological, nutritional, and physio-pathological aspects of the bees. This allows the beekeepers to maintain a safe and healthy environment where the bees can thrive. Precision beekeeping provides beekeepers with accurate information, monitors risks to the bees, and improves productivity. However, these systems are very costly and are difficult to produce on a global scale, reducing their ability to be mass distributed. These systems only focus on domesticated bees while neglecting other wild pollinator species such as butterflies, hummingbirds, and bumblebees. The systems also do not integrate environmental factors such as pesticide exposure, habitat quality, and weather changes, hence no predictive analysis can be conducted.

4.3 World bee Project[14]

The world bee project - remotely monitors bees using artificial intelligence and acoustic technology in small scale farms to better understand bees, monitor health of bees using a scale, monitoring temperature in broods nest, and microphone at the front of the hive, information is gathered on local weather patterns, diseases, parasites, predator species and pesticides, which, when compared with the data from the bees, can give insight into what environmental factors might be impacting the health of the bees.

The World bee project is an initiative that aims to monitor and protect bee populations. It utilizes the deployment of smart hives equipped with sensors similar to precision beekeeping the collected data helps researches understand the effects of climate change, habitat loss, and pesticides on bees. The project focuses

[13] https://pmc.ncbi.nlm.nih.gov/articles/PMC10778344/
[14] https://www.sciencedirect.com/science/article/pii/S2468265922000166

solely on bees, and much like precision beekeeping, it fails to monitor other wild pollinators which also play an important role in the pollination of flowering plants.

Additionally, the project depends on funding from corporations such as Oracle, limiting the focus of the project to objectives which align with the corporate's interest, potentially sidelining broader goals of conservation. Finally predicting the impact of climate change on pollinator populations is a highly complex task to the multitude of variables and factors that can affect pollinator populations climate models often fail to account for localized ecological dynamics, which is integral for predicting pollinator populations.

Although each of these solutions aim to understand the effects of climate change on managed bee populations they fail to monitor other wild pollinating species, and also ignore localized ecological dynamics such as the micro variations in climates between regions, the diversity of different ecosystems, species interactions, and human influence. Our model aims to address the decline in pollinator populations through real-time monitoring, predictive analysis, and intervention planning while addressing the issues faced by preexisting solutions.

5 Methodology

The proposed solution will expand the scope of monitoring by not only focusing on honeybees, but a large range of wild pollinators such as hummingbirds, butterflies, beetles, and bats, allowing us to protect a greater number of species. Furthermore our model will account for Localized Ecological Dynamics through the integration of local climate data, types of ecosystems, and human factors.

Lastly our model will offer real time insights into the behavior of pollinator populations, monitor any microclimate impacts, predicting future trends and allowing for more focused conservation. Data collections that our model requires are :

5.1 Wild pollinator Data

Species identification and tracking - The global pollinator monitoring watch[15] is a large dataset that includes data on species populations, distribution and activity.It can be utilized to identify species in different ecosystems and along with the use of image and video data from camera traps and drones to track different pollinator species, the dataset can be expanded.

5.2 Ecological Data

Data will be collected on interactions between various pollinators and plants. The TRY Plant Database[16] contains an extensive amount of data on the specific interactions between plants and their pollinators. This data set will be mostly sufficient for the model due to its large size.

5.3 Behavioral Data

The activity and movement patterns of wild pollinators can be captured through the use of GPS trackers, drones, and motion cameras. These can be utilized in conjunction with existing databases such as Movebank, Bat monitoring database, monarch watch tagging program, and the global tagging of initiators, an initiative that uses micro GPS tags to collect movement and behavior data of wild pollinators.

5.4 Localized Ecological Dynamics Data

5.4.1 Microclimate Data

Local variations in wind speed, temperature, wind speed, soil moisture obtained from ground based weather stations, and iot sensors which will be deployed in various ecosystems such as urban, rural forest, and agriculture. Data sets like ERA5, Soilgrids, Weather UndergroundAPI, COSMOS Soil Moisture Network, and MODIS (Moderate Resolution Imaging Spectroradiometer) can be utilized to form a baseline set of microclimate data. Pollinators and flowering plants are highly sensitive to climatic conditions such as temperature, humidity, hence it will affect their ability to survive and reproduce.

5.4.2 Habitat Diversity And Ecosystem Health

Data such as floral species diversity, land type (forest, agriculture, urban) along with pesticide use will help us understand how each habitat's land use patterns, human activities, and pesticide use affect pollinators. We will again insight into which types of habitats support pollinator growth and which ones are at risk.

[15] https://www.inaturalist.org/projects/global-pollinator-watch
[16] https://www.try-db.org/TryWeb/Home.php

5.4.3 Species Interactions

As pollinators are part of an ecosystem they are affected by other pollinators, flowering plants, predators, and even competitors. Understanding the dynamics between these populations will give us insight on how we can protect the pollinators. We will be able track any changes that occur within these ecosystems, and detect imbalances or the loss of key species. We can also prioritize conserving the pollinators which play a more vital role in maintaining ecological balance.

5.5 AI/ML Techniques

Once we have collected and fine tuned the data, we will utilize specific AI/ML models to combine microclimatic, habitat, and species interaction data in order to forecast future trends in the populations of pollinators and biodiversity in various ecosystems. We will also be able to identify regions, and species that are at the most risk, in order to focus our conservation efforts and increase efficiency while guiding agricultural practices, climate policies and generating insights that can be applied through global conservation strategies. Due to the sheer size and complexity of the dataset, and scale, we will utilize various ai/ml techniques to tackle species identification, real time monitoring, localized ecological dynamics modelling, and ecological health tracking.

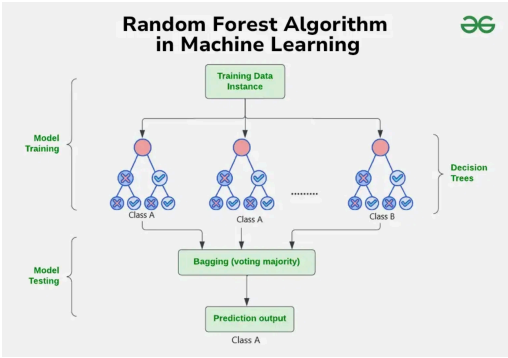
5.5.1 Species identification and classification

Convolutional Neural networks will be utilized to analyze image and video data to extract various features of pollinators such as body morphology, color, patterns, and markings that are unique to different pollinating species. Transfer learning will also be used to improve the Accuracy of the CNN's, which will be fine tuned with existing datasets. Thus provides us with foundational data of various species enabling the understanding of species interactions, behaviors, and even threats.

5.5.2 Real time pollinator monitoring

Through the use of random forests, and support vector machines can be used to classify the health of the pollinators based on certain features such as environmental conditions, pesticide exposure, and even species diversity. Both RF and SVM algorithm can handle complex classification tasks with a large range of features.Due to the non-linear relationship of how multiple environmental conditions can affect pollinator health, and the varying importance of each feature Random Forest is a suitable algorithm. Furthermore, SVM algorithm can be applied to create accurate boundaries between categories of pollinator health even in cases where data is not as well defined.

ARIMA (Autogressive Integrated Integrated Moving Average) and LSTM[17] (Long short term memory) can predict future population trends by analyzing temporal data. Temporal data refers to data that captures information about time and is utilized for analyzing tends and patterns. Data like historical pollinator population counts, climate metrics over time, and changes in land usage can be fed to the system to predict future populations of pollinators. This research paper[18] utilizes the ARIMA model to forecast honey bee populations in Chhattisgarh.LSTM is a type of RNN (recurrent neural network) that can handle sequential data, allowing us to effectively model the relationship between evolving climate conditions and pollinators over long periods of time.The combination of ARIMA and LSTM will ensure that the model covers both simple and complex dynamics within the vast dataset.

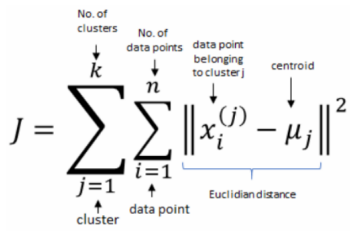


5.5.3 Localized Ecological Dynamics Modeling

K-means[19] is a type of spatial-temporal clustering algorithm that can used to group ecological regions with similar features, thus revealing areas with optimal conditions for pollinators. K-means clustering partitions data into K number of clusters by minimizing the distance between different data points and their centroids(mean of the data set). By finding regions with similarities in environmental conditions, we can

[17] "https://www.bioinf.jku.at/publications/older/2604.pdf"
[18] "https://www.mathsjournal.com/pdf/2023/vol8issue5S/PartC/S-8-4-147-581.pdf"
[19] "https://www.geeksforgeeks.org/k-means-clustering-introduction/""

effectively allocate resources like habitat restoration in regions with suboptimal pollinator health. Additionally, this approach enables us to analyze why certain regions are more suitable for pollinators, providing insights into creating pollinator-friendly habitats in the future.



`figure` - K-means Clustering[20]

5.5.4 Spatial Temporal Models

These models analyze both spatial and temporal data to make predictions. First, microclimatic data, pollinator interaction data, and human activity data will be combined. GIS (Geographic Information Systems) will then be utilized to map these variables globally over a period of time. Then, the models will utilize techniques like Gaussian Processes and Markov Models to predict changes in these parameters. The model will successfully predict how these areas will change over time.

5.5.4.1 Reinforcement Learning

A type of machine learning wherein an agent interacts with an environment and receives feedback, allowing it to learn an optimal action.

5.5.4.2 Key Components

- **Agent:** Algorithm that decides strategies.
- **Environment:** The entire ecological system (pollinators, plants, biodiversity, and human influences).
- **Actions:** Possible interventions such as reducing the use of pesticides and restoring habitats.
- **Rewards:** Actions that improve pollinator populations and the overall health of the ecosystem will be given positive feedback.

5.5.5 Learning Process

The agent will try out different possible actions in a simulated environment, and the outcome will be observed. Over a period of time, the model will learn the most effective actions through positive reinforcement. The model allows us to simulate various strategies and measure their effectiveness, enabling us to take the best course of action.

5.5.6 How These Models Work Together

5.5.6.1 Data Grouping

- **CNNs:** Used to analyze images and videos to identify pollinator species in different regions.
- **RF and SVM:** Predict population health and identify trends.
- **K-means** and **DBSCAN:** Organize both spatial and temporal data into different clusters, highlighting areas with shared characteristics.
- **ARIMA:** Predicts long-term trends in pollinator population dynamics.

5.5.6.2 Localized Predictions

The clustered data will be fed into the spatial-temporal models to provide predictions about the dynamics between each ecosystem, along with identifying risks in different regions.

5.5.6.3 Intervention Planning

Lastly, reinforcement learning will utilize insights from the previous models and recommend the most effective action.

Implementation plan overview

- Data collection : All the data will be collected from the datasets mentioned above, along with drones, sensors, motion cameras, and satellite imagery.
- Data processing de-noised, and stored in a database platform such as AWS or Google Cloud.Python libraries such as Pandas, NumPys, and opencv will be utiulized to clean the data dn image processing.
- Model Development : The AI models will be trained accordingly and tested.

Tools

- CNNs: TensorFlow, PyTorch for image recognition.
- RF/SVM: Scikit-learn for classification tasks.
- ARIMA/LSTM: Statsmodels, Keras for time-series forecasting.
- K-Means/DBSCAN: Scikit-learn for clustering.

[20] "https://linguisticmaz.medium.com/k-means-clustering-clearly-explained-4b61d013ed46"

- Reinforcement Learning: OpenAI Gym for scenario simulations using tensorflow, pytorch, scikit learn, keras and open ai gym

Monitoring

- The models will be updated with data and the perfermonce of the system will be monitored.
- Global scaling Docker and kubernetes will be tulized to deploy the systems.

6 Expected Outcomes

6.1 Environmental Impacts :

Our project can improve the environment by restoring pollinator populations, which are essential for the reproduction of 90% of wild flowering plants. For example, species like bees and butterflies play a critical role in supporting biodiversity by enabling plants to thrive. Through real-time monitoring and AI-driven interventions, our model addresses mismatches caused by climate change, such as early or delayed blooming cycles, ensuring pollinators and plants remain in sync.

By protecting and restoring habitats, such as fragmented forests or urban green spaces, and focusing on at-risk species like the Western Monarch butterfly[21], our approach directly enhances ecosystem health. These efforts help stabilize food webs, promote biodiversity, and build resilience against future environmental challenges, creating a healthier and more sustainable environment for generations to come.

6.2 Economic Impacts :

Our project ensures the stability of pollinator populations, which are essential for 75% of global food crops[22] like apples, strawberries, and nuts. By enhancing pollinator health and activity through real-time monitoring and habitat restoration, we can prevent production declines that would threaten food availability and drive-up costs for consumers (Food and Agriculture Organization)[23].

Unlike artificial alternatives such as RoboBees[24], which are costly and inefficient, our project focuses on preserving natural pollinators that provide these critical services sustainably. This not only supports farmers by ensuring consistent crop yields but also benefits consumers by keeping nutritious foods affordable and accessible. By protecting pollinators, we're safeguarding the global food supply and contributing to a more resilient agricultural system.

6.3 Social Impacts :

The decline in pollinators poses a significant risk to global food security, as essential crops rely on pollination for consistent yield and quality. Studies show that strawberries and apples pollinated by insects tend to be larger, more uniform, and better tasting; (Entomology)[25] (PLOS Journals)[26].

Ensuring healthy pollinator populations will help maintain the affordability and accessibility of these crops. Public awareness initiatives, such as the World Bee Project[27], have demonstrated that AI-driven tools can raise public awareness and engage communities in conservation efforts.

Expanding these efforts to include wild pollinators like hummingbirds and bats will amplify conservation outcomes. Additionally, the project will create opportunities in fields like AI development, environmental research, and policy-making, fostering growth and innovation.

6.4 Performance Metrics

6.4.0.1 Pollinator Health and Diversity :

Tools like the Bumblebee Conservation[28] Trust's dataset will track species distribution and abundance, providing essential insights into pollinator population trends and their responses to environmental changes, guiding conservation.

[21] "https://wildlife.ca.gov/Conservation/Invertebrates/Monarch-Butterfly"
[22] "https://ourworldindata.org/pollinator-dependence"
[23] "https://www.fao.org"
[24] "https://thebiologist.rsb.org.uk/biologist-features/robobees-2"
[25] "https://entomology.ces.ncsu.edu/small-fruit-insect-biology-management/strawberry-pollination-basics/?utm_source=chatgpt.com"
[26] "https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0153889"
[27] "https://worldbeeproject.org/"
[28] "https://www.bumblebeeconservation.org/what-we-do/our-projects/current-projects/trust-led-projects/beewalk/"
[29] "https://www.usda.gov/pollinators"

6.4.0.2 Ecosystem Stability:

Metrics such as plant reproduction rates and biodiversity indices will assess ecosystem health. USDA's Pollinator Data will be utilized to measure pollinator activity and its impact on ecosystems, ensuring that interventions align with long-term stability goals; (USDA)[29].

6.4.0.3 AI Model Effectiveness:

Advanced AI models such as CNNs (for species identification), ARIMA, and LSTM (for population trend predictions) will be evaluated for their accuracy and precision. CNNs have been successfully applied in ecological studies for species identification, as highlighted in the article "Deep learning for plant identification: A review[30]."

Time-series models like ARIMA and LSTM, pivotal for forecasting ecological data trends, are detailed in the article "Comparative Performance Analysis of Deep Learning, Classical, and Hybrid Time Series Models in Ecological Footprint Forecasting[31]." These tools will provide real-time insights into pollinator health and population behavior, enabling proactive conservation efforts.

6.5 Impact Assessments

6.5.1 Environmental Impact:

The project's success in restoring biodiversity will be measured using biodiversity indices, habitat restoration metrics, and reductions in pollinator mortality rates. (PLOS Journals)[32]. Real-time modeling based on IPCC climate data will support adaptive interventions by identifying ecological strategies that effectively mitigate pollinator decline and its cascading effects on ecosystems.

6.5.2 Economic Impact:

Improved agricultural productivity and stabilized crop prices will demonstrate the project's economic benefits. Cost-benefit analyses will evaluate the viability of natural restoration methods compared to artificial solutions like RoboBees, which remain economically infeasible due to high development and maintenance costs.

6.5.3 Social Impact:

Surveys and the adoption rates of eco-friendly agricultural practices will measure social success. Community engagement initiatives like the World Bee Project have shown how collaborative efforts raise awareness and foster conservation outcomes. Expanding these initiatives will strengthen public participation and encourage sustainable practices.

7 Challenges and Limitations:

There are a lot of challenges that we have to deal with to bring our work into action successfully.

- **Outdated data:** Existing datasets are region-specific, outdated, or focused only on certain species as we have very limited data on wild pollinator species like butterflies, bats, and beetles.
- **Processing problems:** Combining diverse and large amounts of data sources requires significant processing as the data that we get or is available on the net are in various formats which will cause problems like formatting and understanding data.
- **Computational complexity:** The problem we are trying to target has high computational requirements for training convolutional neural networks (CNNs), running random forests (RF), support vector machines (SVMs), clustering algorithms (K-Means, DBSCAN), and ARIMA models.
- **Field data collection:** There will be a lot of challenges in deploying IoT sensors, and GPS trackers due to cost and maintaining them in different environmental conditions.

7.1 Limitations of Proposed Models:

- **CNNs:** The models like CNNs require large amounts of data for accurate and better calculations which may not always be available. It is pretty hard to train this model if the data is small or unbalanced.
- **RF/SVM:** Random Forests normally struggle with high dimensional data and SVMs are computationally expensive for large datasets.
- **K-Means/DBSCAN:** K-Means is a sensitive model (Sensitive to initial conditions) and it needs a lot

[30] "https://www.sciencedirect.com/science/article/abs/pii/S016816992300460X"
[31] "https://www.mdpi.com/2076-3417/14/4/1479"
[32] "https://journals.plos.org/plosbiology/article?id=10.1371%2Fjournal.pbio.0020383"

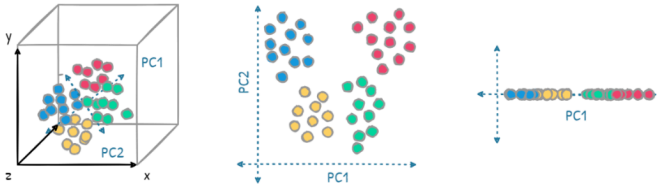
of prior specifications and data while DBSCAN can misclassify points in datasets with varying density.

- **ARIMA:** This model is effective only with linear data and struggles with complex and non-linear ecological data.

7.2 Strategies to Mitigate Challenges

- **Enhancing data availability :** Partnering up with global initiatives like Global Pollinator Watch[33] and TRY Plant Database[34] to supplement datasets. Building employs citizen science platforms and community involvement to collect data.To deal with the problem with CNNs we can use transfer learning and augmented datasets with synthetic data so that we minimize the scope of error and get better outputs. Applying transfer learning for CNNs to reduce training times and computational costs.

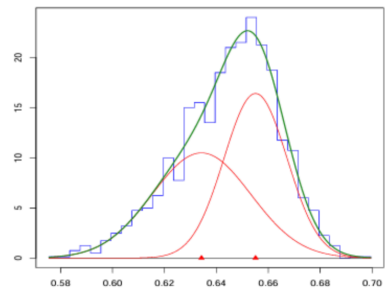
- **Improving Data Integration :** Utilizing AI-ML tools to process, format, and validate different datasets developing a universal dataset format so that our model can understand better are some strategies.Optimize hyper-parameters (best optimal parameters) and use dimensionality reduction techniques like PCA to improve the efficiency of RF/SVMS can be done.



`figure` - Dimensionality Reduction

- **Optimizing Computational Efficiency:**

Usage of cloud-based solutions (ex. AWS, Google Cloud) to manage high computational demands. Usage of advanced clustering methods like Gaussian Mixture Models or hierarchical Clustering is needed to enhance K-Means/DBSCAN models.



- Long-Term Predictive Modeling:

Utilize ARIMA and LSTM models for better outputs, and time-series forecasting of pollinator populations. Integrate climate models (e.g., ERA5, MODIS) to account for long-term environmental changes. Combining ARIMA with non-linear models like LSTM to capture complex patterns in time-series data will make the predictive models even more accurate.

7.3 Effectiveness over 10 years:

Over the next 10 years with advancements in AI, there will be improved pollinator monitoring and conservation. There will be more data available as time passes and the advancement in technologies like IoT sensors and drones will enhance the coverage. Predictive AI models will help make policies for agriculture so that agriculturists adopt pollinator-friendly practices. Clustered and predictive data will allow us to understand more about the pollinators and take action quickly if any species is in danger or might be in danger. The insights gained from our approach can lead to the development of new alternative technologies such as robotic pollinators (as mentioned above). While robotic pollinators are practically impractical but this dataset and AI model can help in the functionality and development of robotic pollinators. The advancements in open-source tools, cloud computing, and cheaper hardware will make this model more effective to use.

This model and the data from this will help raise global awareness and change a lot of policies so there will be a safer future for pollinators. This will even help the ecosystem and us to adapt to climate change. Most importantly this data and model can help prevent pollinator population collapse which could save billions in agricultural revenue and ensure food supply stability.

[33] "https://www.inaturalist.org/projects/global-pollinator-watch"
[34] "https://www.try-db.org/TryWeb/Home.php"

Contribution Table

Name	Roll Number	Contribution	Contribution Details
Tharun Varma D	SE23UARI137	YES	Report writing, research and data collection
L Joy Nirantar	SE23UARI169	YES	Report writing, research and data collection
Sahasri Gundapuneedi	SE23UCAM019	YES	Report formatting, Research, Proof reader
Gatri Reddy	SE23UARI109	YES	PPT presentation
Meher Aditya Varma	SE23UECM070	YES	Data Collection
Pandrangi Varun	SE23UCSE137	YES	Research and Report writting
Pranishk Reddy	SE23UARI099	YES	Image collection
Pranav Reddy	SE23UARI025	YES	PPT and report writting
Pranaya Prabhas Inala	SE23UMEE0023	YES	Grammar Checker
Shradha Nambiar	SE23UMEC027	YES	Spelling Checker